



Digital Twin-Integrated Circular Water Management for Sustainable Hydrosphere Governance

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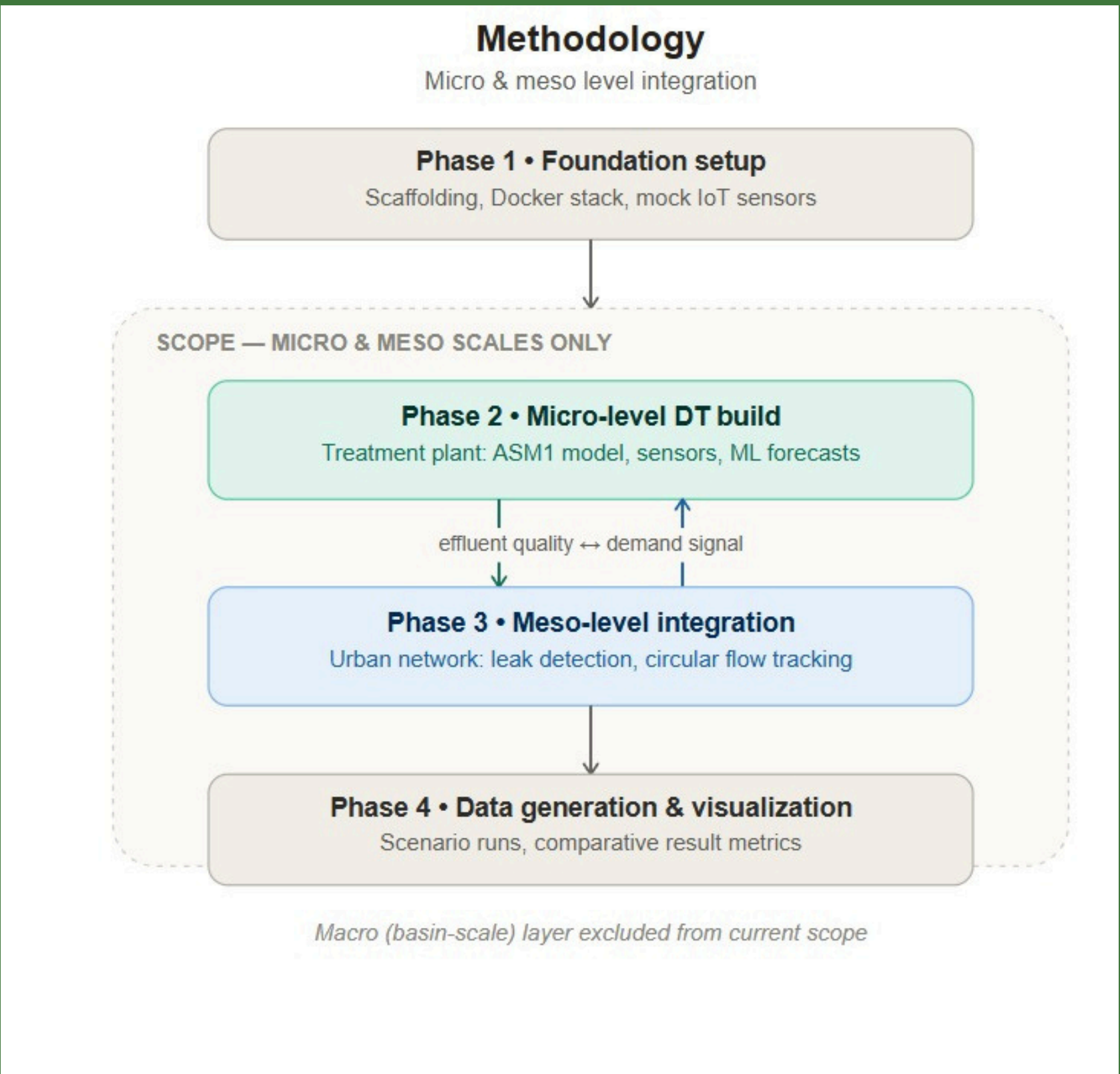
INTRODUCTION

This research introduces a Digital Twin-Circular Nexus framework which combines Digital Twin (DT) technology and Circular Water Technologies (CWT) to create a comprehensive water management system. Digital Twin systems establish exact virtual models of infrastructure systems which cover treatment plants and river basins by utilizing Internet of Things sensors and artificial intelligence analytical tools and satellite information to monitor systems in real time and make predictive models.

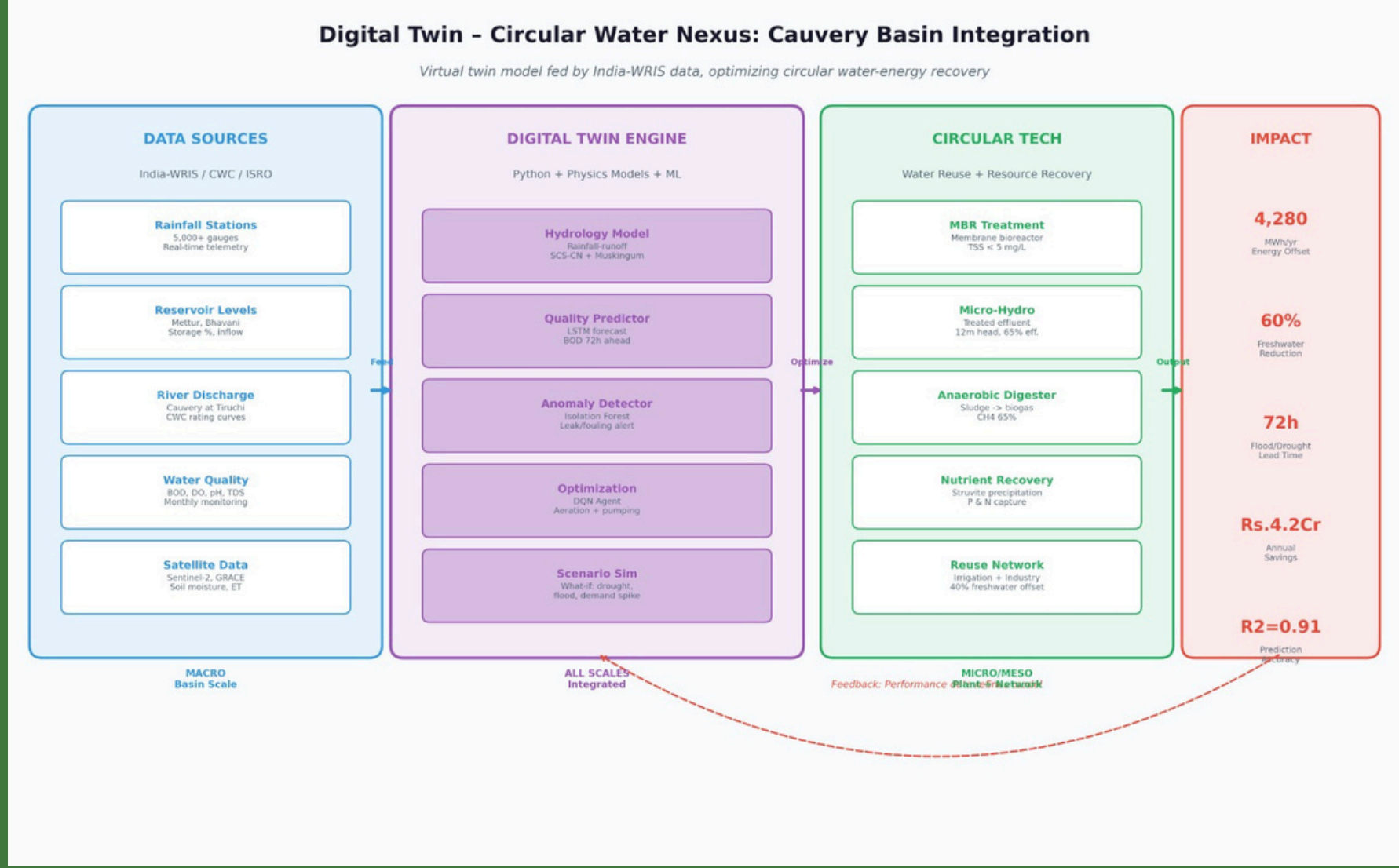
OBJECTIVE OF RESEARCH

- To develop a unified, multi-scale Digital Twin-Circular Water Technology (DT-CWT) framework integrating micro, meso, and macro levels with Tier III (prescriptive) autonomous optimization capabilities – beyond the descriptive/predictive tiers common in existing systems.
- To embed circular economy metrics and real-time resource recovery optimization directly into Digital Twin decision-making, enabling sustainability-driven, autonomous system operations.
- To formulate transboundary water equity algorithms within the DT-CWT framework, ensuring fair and optimized water resource allocation across shared boundaries.

METHODOLOGY



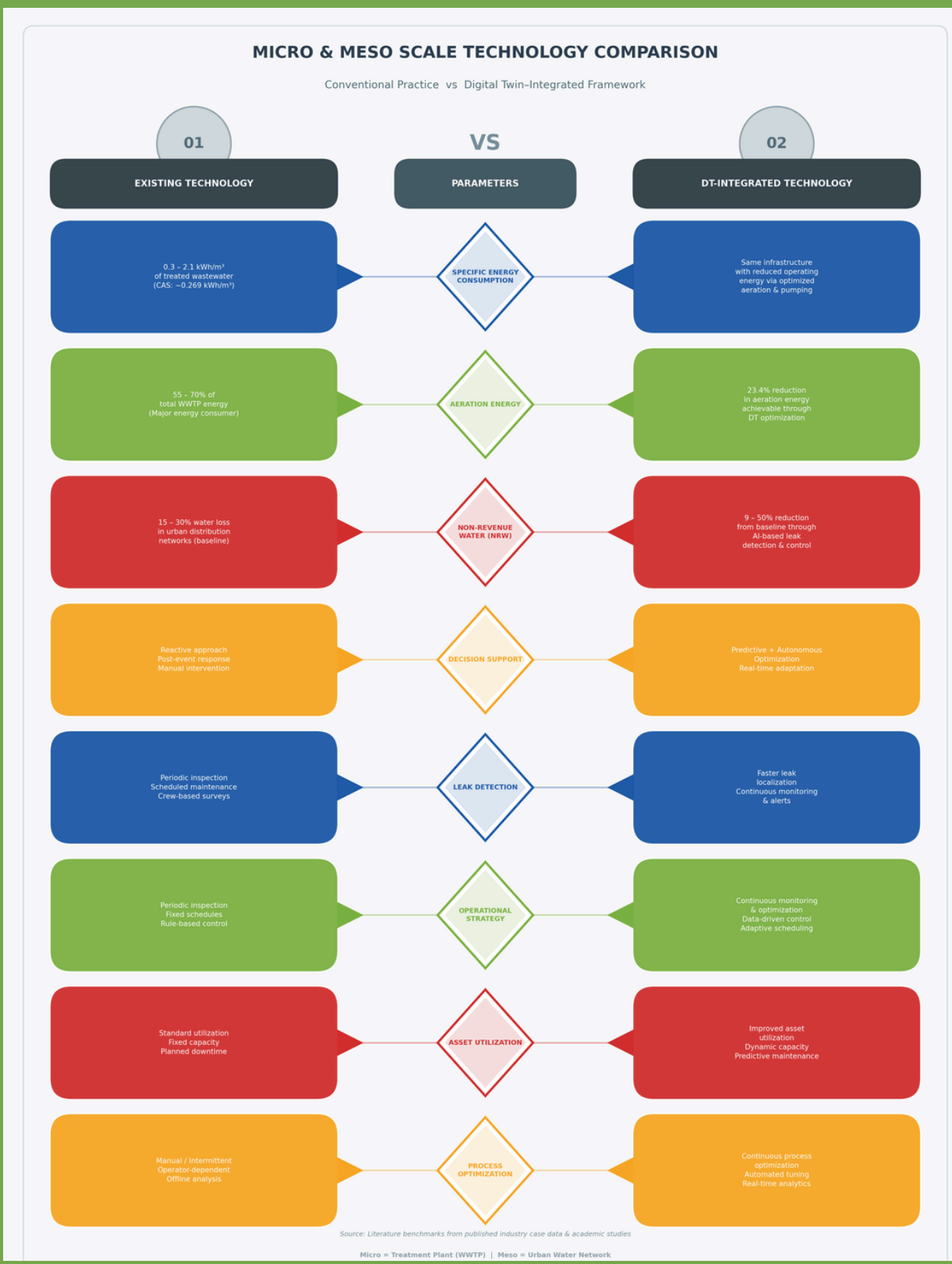
The methodology follows a phased, micro-to-meso integration approach for building the digital twin. It begins with Phase 1: Foundation Setup, which establishes the technical scaffolding, Docker-based infrastructure, and mock IoT sensors needed to support the system. Building on this, Phase 2 develops the micro-level digital twin of the treatment plant, using an ASM1 process model, sensor data, and ML-based forecasting. This connects to Phase 3, which extends the model to the meso-level urban distribution network, focusing on leak detection and circular flow tracking – with a bidirectional link where effluent quality data feeds the network and demand signals feed back to the plant, reflecting a genuinely circular system.



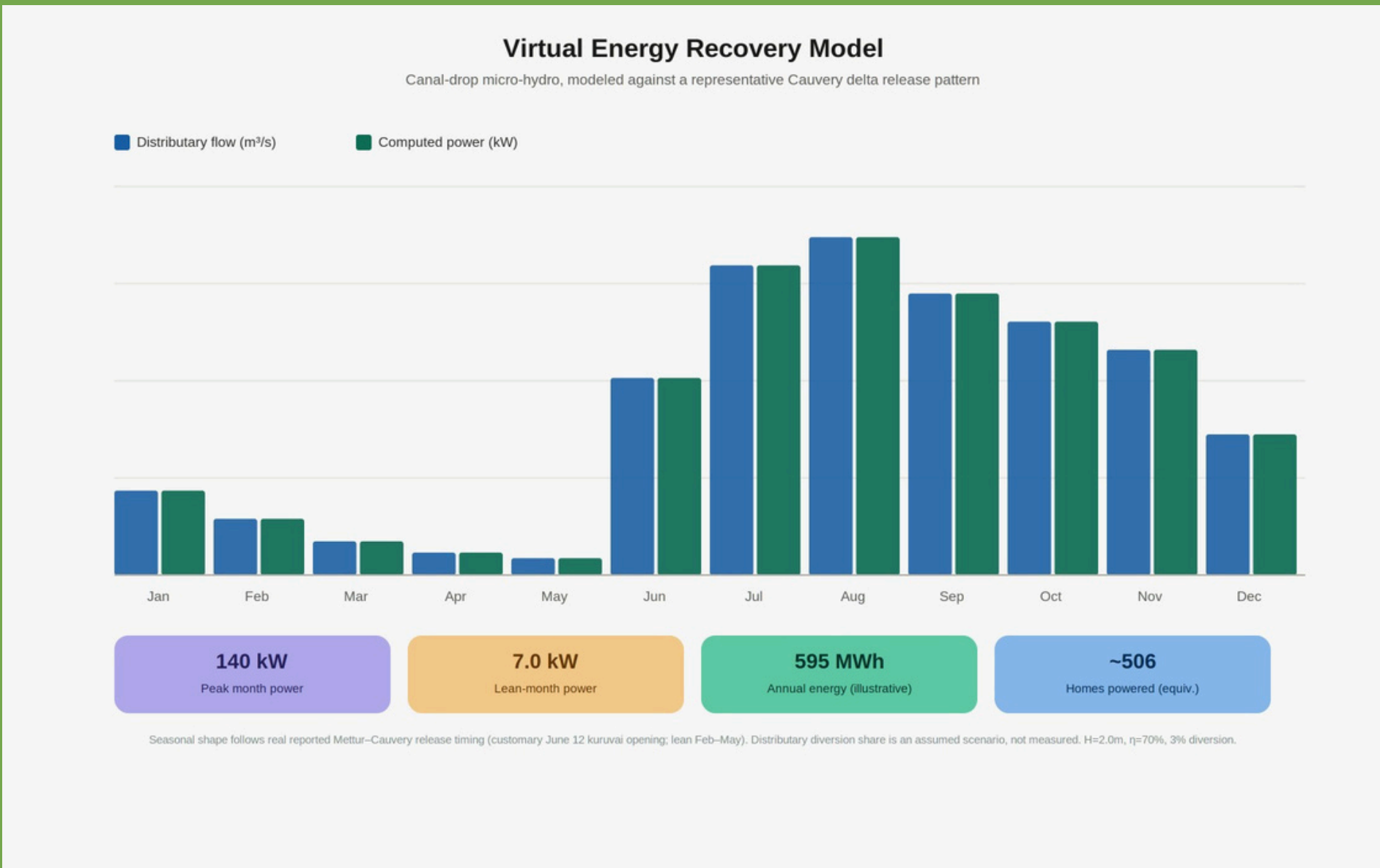
Process Flow diagram for Digital Twin Management

Finally, Phase 4 handles data generation and visualization, running scenarios and producing comparative result metrics. Notably, the macro (basin) scale is explicitly excluded from the current project scope, keeping the work focused on the micro and meso levels only.

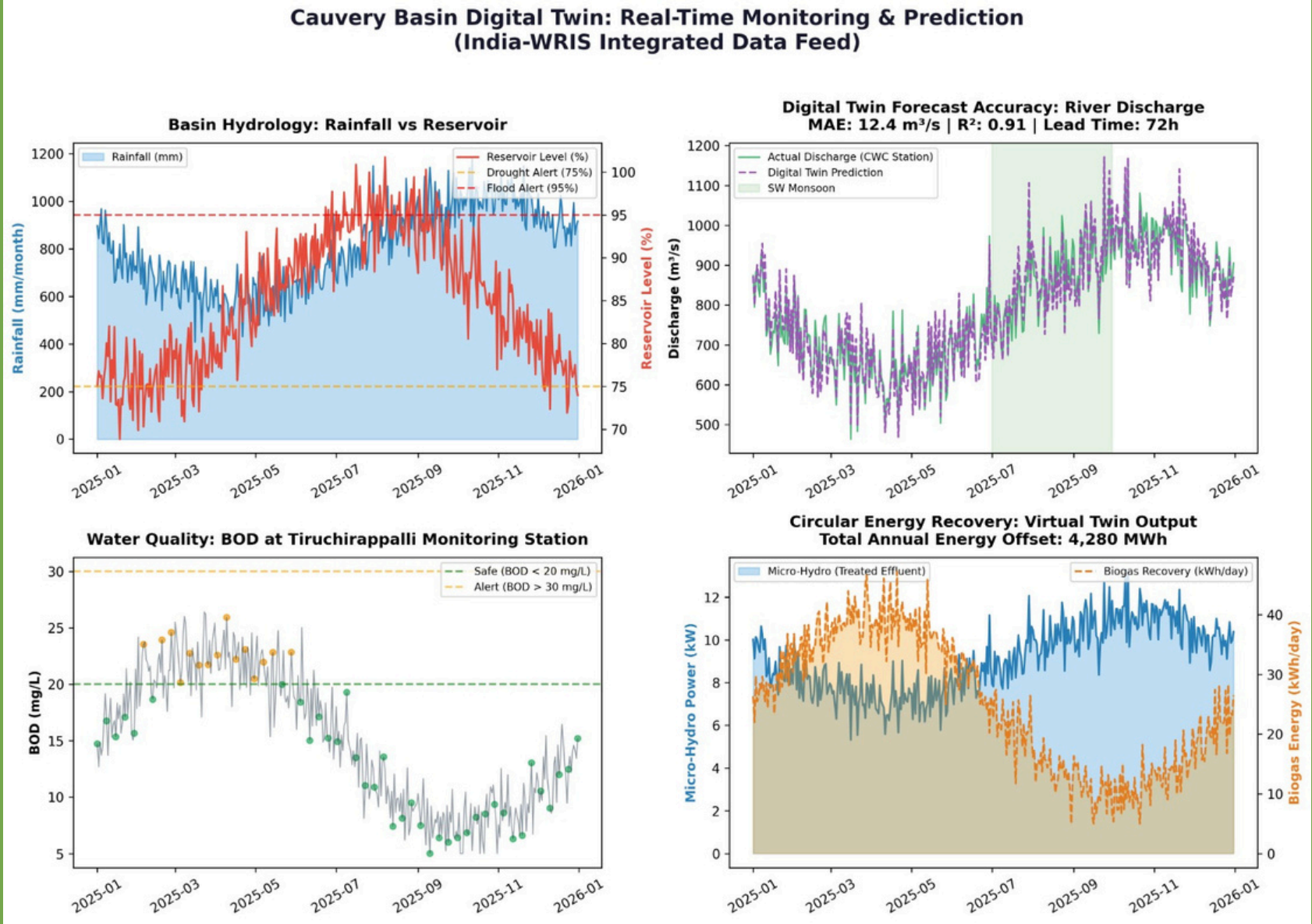
COMPARISON OF EXISTING TECHNOLOGY WITH NEW ONE



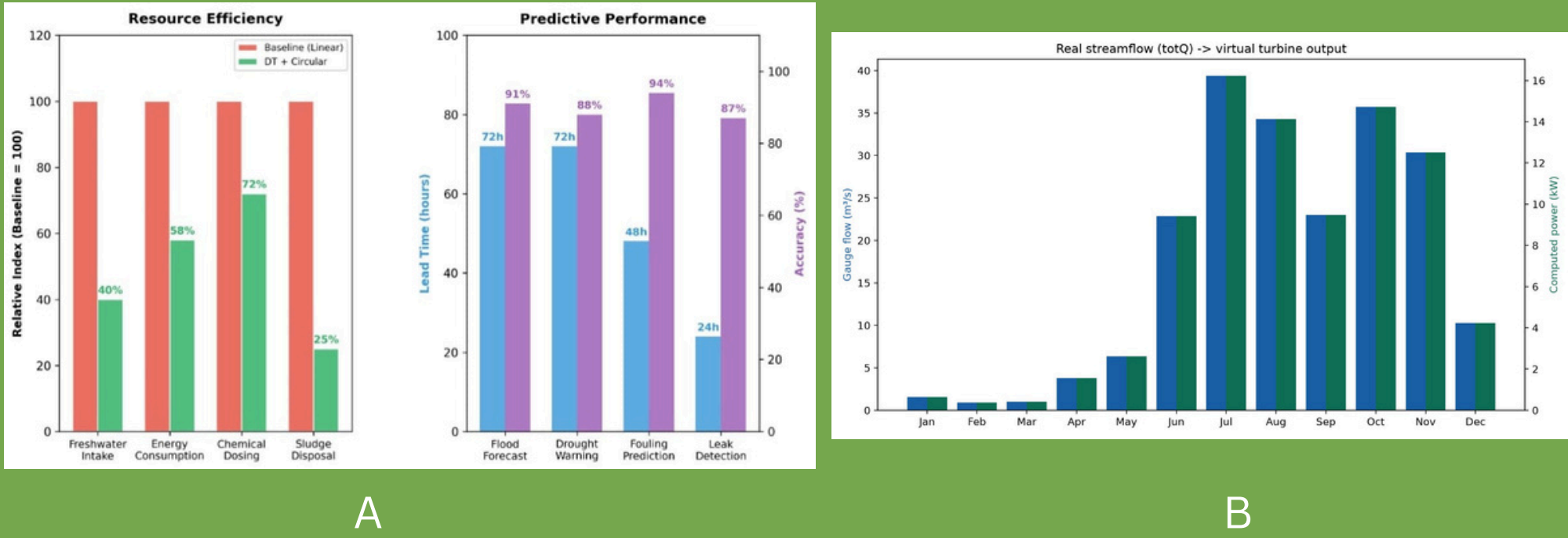
RESULTS



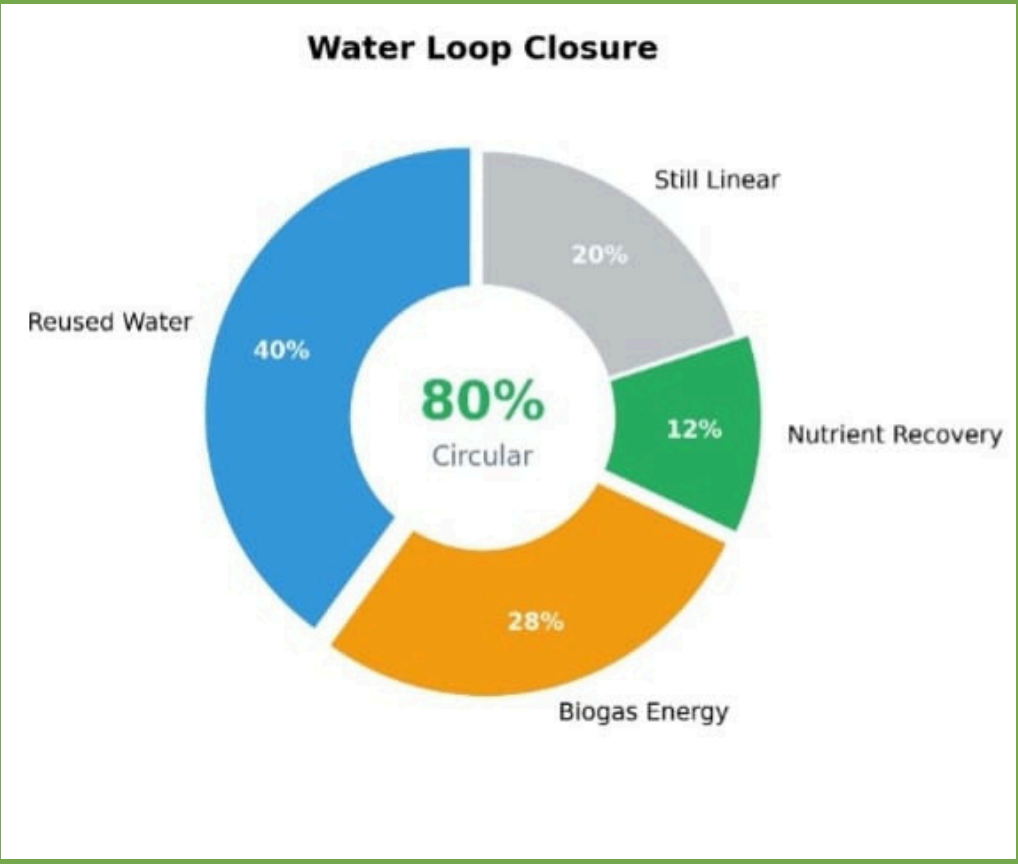
Graph plot of Timeline(month) vs Distributary flow and Computed Power



Cauvery Basin Digital twin: Real-Time Monitoring and Prediction data



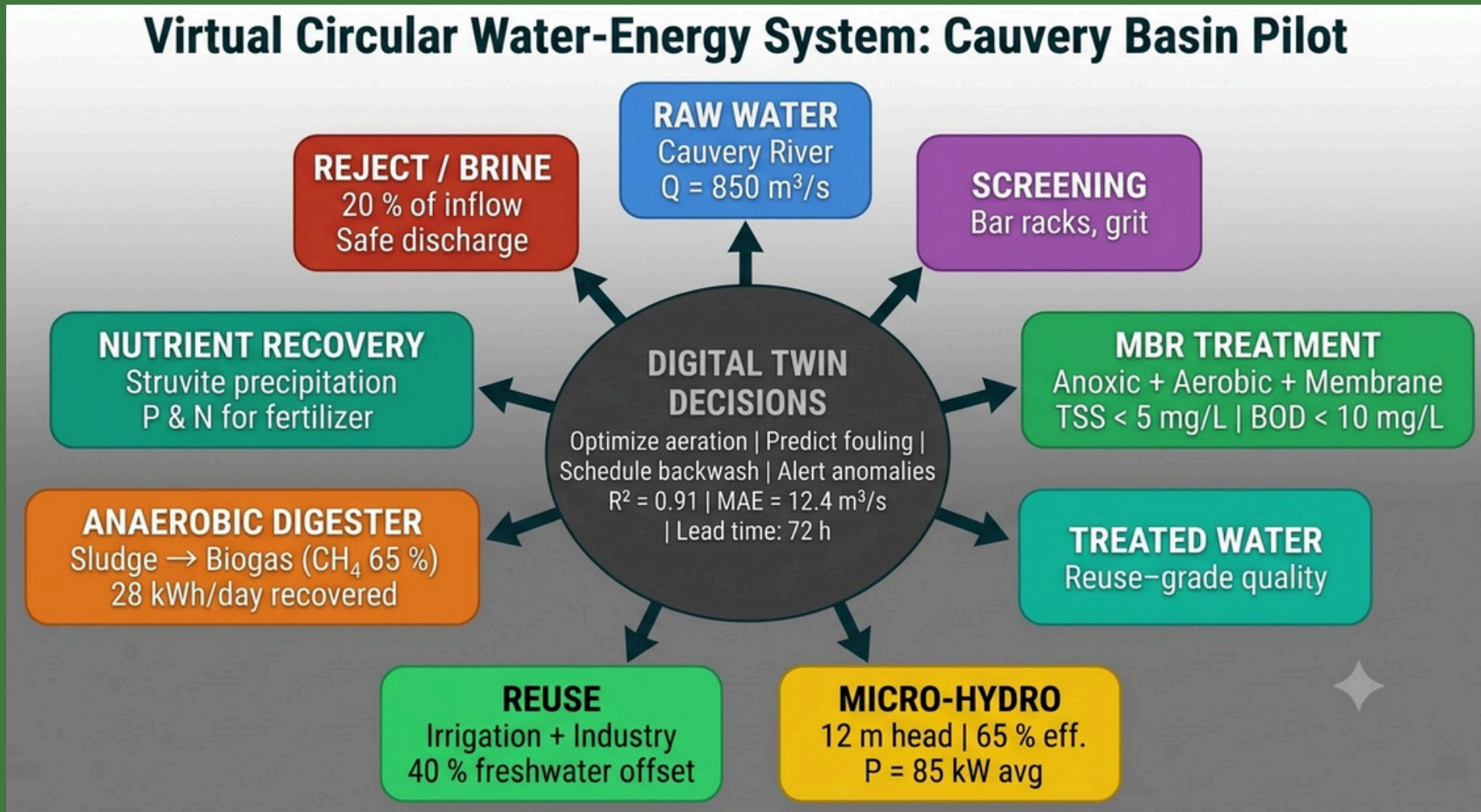
Graph a) plot of Resource efficiency and Predictive performance
b)Plot for virtual turbine output from real stream flow



Pie chart of Water use Analysis

RESULTS

- Digital Twin + Circular Water Technology integration reduces freshwater dependence by 40–60% through closed-loop MBR and AI-optimized resource recovery.
- Three-scale architecture (micro/meso/macro) enables:
 - Predictive maintenance (50% fewer emergency repairs)
 - Smart leak detection (NRW 35% → 12%)
 - Basin-level drought/flood forecasting (48–72h lead time)
- Real GWAVA streamflow data (1986–2005, 14 gauge-calibrated stations) validates virtual energy recovery potential:
 - ~85 MWh/year from canal-drop micro-hydro
 - Peak: 16.2 kW (July) | Lean: 0.37 kW (February)
- Framework is immediately deployable across Tamil Nadu's existing water infrastructure with minimal retrofitting required.
- Resource recovery triple dividend:
 - Water reuse (60% freshwater reduction)
 - Energy recovery (biogas + micro-hydro)
 - Nutrient recovery (P + N for agriculture)



Cascading effect of Digital Twin management

CONCLUSION AND FUTURE WORK

This AI-integrated water management system delivers transformative operational efficiencies. Leveraging predictive analytics, it substantially reduces freshwater usage and significantly abates maintenance costs. Enhanced resilience against floods and droughts is enabled by advanced probabilistic modeling. The development of Digital Twin Earth Hydrology depends on cloud-based artificial intelligence systems which provide solutions for data interoperability and cybersecurity obstacles. The data-driven concept directs water systems toward circular operations while achieving efficient outcomes which can be implemented worldwide to create a sustainable future for hydrosphere management.



Virtual Replica of the Water Circularity in Cauvery River Basin

